

## Group Assignment - Jupyter Notebook, Google Colab & Libraries for Data Science

### Business Context

A Portuguese banking institution conducted direct marketing campaigns by telephone to offer customers a term deposit. The bank would like to understand which customers are more likely to subscribe and whether Machine Learning can help identify promising customers for future campaigns.

### Dataset

Use the UCI Bank Marketing dataset ("bank-full.csv", link below). It contains 45,211 observations with a mixture of numerical and categorical variables. The column y: 'yes' if the customer subscribed to the term deposit and 'no' otherwise. The dataset includes customer characteristics, financial information, campaign information and previous contact history.

Dataset source: [UCI Machine Learning Repository - Bank Marketing](#)

### 1. Explore the data

Perform an exploratory analysis and identify relevant patterns related to successful subscriptions.

### 2. Prepare the data

Prepare the numerical and categorical variables for Machine Learning and decide which variables should be included or excluded.

### 3. Build a baseline model

Create a train/test split and train a Logistic Regression model. Evaluate the result using suitable classification metrics such as precision, recall, F1 score.

### 4. Compare different models

Train at least two additional scikit-learn classification models and compare them with the baseline.

### 5. Feature engineering

Create at least two additional features that you believe could improve the prediction and explain your reasoning.

### 6. Interpret the results

Summarize the most important findings, compare the models and explain how the bank could use the results in a future marketing campaign.

### Documentation and AI Usage

Study the official documentation of the packages used in the course and explore functions that were not demonstrated during the lecture. The use of AI tools is explicitly allowed for understanding documentation,

generating or debugging code, suggesting features or interpreting results. However, the group must critically review the generated code and be able to explain the chosen approach and results.

**Deliverable**

Submit one **Jupyter Notebook (.ipynb)** per group containing executable code, relevant tables and visualizations, model comparisons, short Markdown explanations and a final business-oriented conclusion.